Beyond a Single Scale of Intelligence

Likun Xie

Much of the concern around AI depends on what we believe it will eventually become. This matters not only emotionally but practically: any discussion of how mathematics should respond to AI depends, at least partly, on some view of its future abilities.

This essay is mostly about the individual and personal side of these questions: how it feels to do mathematics, what we value in it, and what continues to motivate us. I am not trying to address the broader practical, social, or institutional consequences of AI in mathematics. Nor do I think there can be a single answer that applies to everyone. Different people value different aspects of mathematics, work in different ways, and are motivated by very different things. What I write here is simply one way of thinking about these changes from my own perspective.

One thing that seems important to me is that intelligence, understanding, and styles of thinking do not lie on a single measurable scale. Being much stronger in one kind of intellectual task does not necessarily mean being stronger in every other kind.

Robotics offers a simple analogy. An AI-powered robotic animal may reproduce many of the functions of a real one: it can walk, respond, and adapt to its environment. But simple movements can remain strangely unconvincing. We may immediately feel that something is too rigid or not quite alive. Even when such movements are effortless for a living animal, machines may struggle to reproduce them naturally. Human sensitivity to these details grows out of perception, embodiment, and experience.

Something similar may be true in mathematics. Certain judgments that come naturally to an experienced mathematician, such as seeing the right formulation, sensing which structure matters, or recognizing that an argument misses the real point, may be difficult to reproduce even when AI's mathematical abilities become extremely powerful.

What AI May Change

A large part of mathematical work seems especially suitable for AI: symbolic manipulation, searching through possible arguments, testing examples, combining techniques, and carrying out long technical proofs. This often makes mathematics a visible benchmark for AI progress. AI may eventually become much more powerful than any individual mathematician at these tasks.

Recent work on Navier–Stokes illustrates this possibility. The progress is genuinely impressive, but it also builds on a substantial body of earlier and concurrent work by mathematicians including Diego Córdoba, Luis Martínez-Zoroa, Tristan Buckmaster, and Levent Alpöge, which had already developed related mechanisms and promising directions. The AI was able to take these ideas, explore alternative possibilities, and push the broader framework much further, carrying out intricate arguments and testing technical possibilities at a scale that would take human mathematicians much longer. What seems especially striking is how effectively AI can advance a promising framework: combining constructions, repairing failed attempts and overcoming barriers that might otherwise require a great deal of time.

So far, the evidence seems much stronger that AI is becoming exceptionally good at technical mathematics and exploration within existing or plausible frameworks than that it is equally good at creating fundamentally new conceptual viewpoints.

But this alone could change the pace of mathematics dramatically. In the past, a major

conceptual breakthrough could generate decades of subsequent work as mathematicians developed its consequences, tested it in different settings, refined the methods, and discovered further applications. If AI becomes extremely good at this kind of development, much of that process could happen far more quickly. A single conceptual advance might generate a large body of meaningful mathematics in a much shorter time.

In that sense, AI would not need to replace conceptual invention in order to have a profound effect on mathematical research. It could greatly accelerate everything that comes after a good idea.

Mathematical Insights

Mathematical research involves more than proving difficult statements. Sometimes a mathematician looks at a theorem and feels that it has been formulated in the wrong way. A proof may be correct but still fail to explain the phenomenon. At other times, one senses that a particular approach is more promising than others, or that existing methods are not enough and a genuinely new mechanism or framework is needed.

Mathematicians often say things like, "There should be a more conceptual explanation," or "There should be a much simpler proof from the right perspective." Such judgments are difficult to formalize. They are not simply reactions to a proof being too long or the hypotheses too complicated; they arise from an internal picture of the subject built through deep mathematical understanding and experience.

That internal picture develops slowly and individually. We may spend years misunderstanding things and pursuing ideas that fail, learning different perspectives, and gradually deepening our understanding. Over time, we develop a personal sense of how a subject behaves, shaped by the problems we have worked on, the mistakes we have made, and the viewpoints that have influenced us. Mathematical intuition is not just a collection of facts, it is also an individual way of organizing them, recognizing what matters, and sensing the structure behind them. Interesting insights often grow out of this kind of deep understanding and experience. There is something profoundly human and individual in this process.

Perhaps AI will develop something comparable. But here I am reminded again of the example of robotics imitating a living animal. A robotic animal may reproduce many behaviors extremely well, yet still struggle to make even a simple movement feel as alive as that of a real animal. Likewise, AI may reproduce many outward features resembling mathematical intuition without developing the same kind of insight and internal sense that grows out of a mathematician's deep and individual understanding of a subject.

AI and Maths Conjectures

(Oct 16, 2026) Recently there have been several AI-generated attempts at major mathematical conjectures, including some that I care about, such as the small Cohen–Macaulay conjecture and Serre's positivity conjecture. We don't know yet whether all of these arguments are correct, and I think they still need serious verification.

I think what makes a conjecture interesting is often the possibility of exploring new mathematical territory, either along the way to a solution or from whatever the solution itself might reveal. A conjecture usually points toward some underlying structure or theory that we are trying to understand. Sometimes a proof reveals that structure clearly, while other times it settles the conjecture without explaining nearly as much. Either way, many questions remain about why

the result is true, whether there is a more conceptual explanation, and what else we can learn from it.

A counterexample can leave even more questions behind, since it usually tells us that something fails without revealing much about the underlying structure or why it fails. This makes it even more interesting to understand what causes the failure and under what conditions the conjecture might still hold. Sometimes a counterexample can also point toward new structures or theories that we had not considered before.

Different Kinds of Intelligence

AI may become much better than humans at exploring possibilities and carrying out complicated proofs within existing frameworks, while humans retain distinctive strengths in mathematical taste and insight, conceptual framing and innovation, and recognizing what matters and which directions are worth pursuing. In mature research, these judgments often determine what happens next.

A future mathematician may say to an AI:

"You have proved something much harder than what I could prove myself, but I think you are still looking at the problem in the wrong way."

The mathematician changes the formulation, introduces a new approach or framework, and the AI then carries out an enormous amount of difficult mathematics within it.

Such a future would look very different from one in which human mathematicians are left defending an ever smaller territory against AI. If one assumes that every human cognitive advantage is temporary, then each breakthrough can seem like evidence that the space for human contribution is shrinking. But that assumption is far from established.

Mathematics may instead develop as a collaboration between different kinds of intelligence: machines extraordinarily powerful at search, technical development, and scale, and humans contributing judgment and insight, conceptual framing and innovation, and mathematical taste grounded in deep understanding of the subjects. Working with AI may simply become one research style among many, shaped by personality and preference. Just as some mathematicians think best through frequent discussion and collaboration while others prefer long periods of solitary work, some may find constant interaction with AI highly productive, while others may use it only occasionally or not at all.

Whatever style one chooses, technical ability and deep understanding will remain essential. Mathematical training will likely remain much the same: we will still learn mathematics deeply, master techniques, and build strong foundations. Effective collaboration with AI itself depends on substantial mathematical skill. One needs enough command of the details to judge, modify, and develop what the AI produces, and to work on problems that AI still cannot solve.

Given that the present and future remain uncertain, there may be no single practical adjustment to make right away. The mathematical community will adapt gradually as new problems and discussions arise, and mathematical research will find a new balance over time. Whatever form these changes take, the mathematical values we have been discussing will remain important. We should continue to enjoy the mathematics we deeply care about, deepen our understanding, follow the questions that matter to us, and become the mathematicians we want to be. These are the things that have always mattered, and they will continue to matter both to our hearts and to the mathematical community.